Name: _____

TOPIC TEST 2

Chapter 6 Applications of integration

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Five multiple-choice questions (5 marks).
- Part B: Six short-answer questions (25 marks).
- Part C: Two analysis questions (10 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 $\int (4x + 7)^3 \, dx =$

A $16\,(4x + 7)^4 + c$

B $\frac{1}{16}(4x + 7)^4 + c$

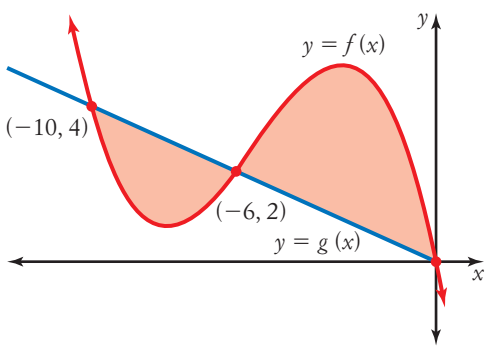
C $(4x + 7)^4 + c$

D $\frac{1}{4}(4x + 7)^4 + c$

E $4(4x + 7)^4 + c$

[1 mark]

2 The shaded area at right can be written as:



A $\int_{-10}^{-6} f(x) - g(x)dx + \int_{-6}^{0} g(x) - f(x)dx$

B $\int_{-10}^{0} f(x) - g(x)dx$

C $\int_{-10}^{-6} g(x) - f(x)dx - \int_{-6}^{0} f(x) - g(x)dx$

D $\int_{-10}^{-6} g(x) - f(x)dx + \int_{-6}^{0} f(x) - g(x)dx$

E $\int_{-10}^{0} g(x) - f(x)dx$

[1 mark]

3 $\int_0^4 (3\sqrt{x} + x)dx$

A 24

B 16

C 32

D 20

E 0

[1 mark]

4 If the derivative of $e^{x^3 + 6x}$ is $3(x^2 + 2)\ e^{x^3+6x}$, then the antiderivative of $(x^2 + 2)\ e^{x^3+6x}$ is:

A $9(x^2 + 2)\ e^{x^3+6x}$

B $\frac{1}{3}(x^2 + 2)\ e^{x^3+6x}$

C $\frac{1}{3}\ e^{x^3+6x} + c$

D $\frac{1}{3}(x^2 + 2)$

E $3(x^2 + 2)\ e^{x^3+6x}$

[1 mark]

5 Water flows into a container at the rate $R'(t) = 10e^{0.2t}$ (L/min) where t is in minutes. What is the total number of litres (to the nearest litre) that flowed into the container in the first 5 minutes?

A 50

B 86

C 120

D 136

E 75

[1 mark]

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Part B

Short-answer questions

Six short-answer questions
25 marks
Show your working where appropriate.

1 Find each indefinite integral.

a $\int (3x^4 + 2x)dx$

b $\int (4 - 3x)^2 dx$

c $\int \frac{3x^2 - 4x + 7}{\sqrt{x}} dx$

[4 marks]

2 Evaluate the following:

a $\int_{-1}^{2}(9x^2 - 6x + 2)dx$

b $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 4\ \sin(3x)dx$

c $\int_{1}^{3}\frac{dx}{(2x+5)^2}$

[6 marks]

3 Find y in terms of x if $\frac{dy}{dx} = 14x - 4$ and $y = 8$ when $x = -1$.

[3 marks]

4 Find the area under the curve $y = 3x^2 + 3x - 18$ from $x = -2$ to $x = 1$.

[5 marks]

5 Find the area enclosed between the curve $y = x^2$ and the line $y = x + 2$.

[4 marks]

6 Fluid flows out of a storage tank at the rate

$F'(t) = 500e^{-0.4t}$ litres/h

where t is measured in hours.

How much fluid will flow out of the tank in the first 5 hours?

[3 marks]

Part C

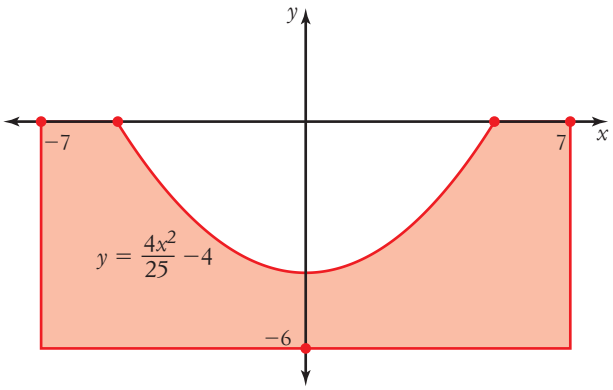
Analysis questions

Two analysis questions
10 marks
Show your working where appropriate.

1 The diagram below represents a cross-section of a concrete culvert. All dimensions are shown in metres.

a What is the area of the cross-section?

b If the culvert is 15 m long, what is the volume of concrete it contains?



[4 marks]

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2 It can be shown that the acceleration of a particle is given by the differential equation $\frac{d^2x}{dt^2} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$, where v is the velocity.

The acceleration of a particle is given by $\frac{d^2x}{dt^2} = 6x^2 - 4x - 3$, where x is its displacement. Find the exact velocity when the particle is 2 cm from the origin if initially the particle is at the origin and has a velocity of 3 cm/s.

[6 marks]

[Total marks: 40]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $\int (4x+7)^3\,dx = \dfrac{(4x+7)^4}{4\times 4} + c$

$= \frac{1}{16}(4x+7)^4$

$\therefore$ **B**

[1 mark]

2 Area between $x = -10$ and $x = -6$ is $\int_{-10}^{-6} g(x) - f(x)\,dx$

Area between $x = -6$ and $x = 0$ is $\int_{-6}^{0} f(x) - g(x)\,dx$

Total area $= \int_{-10}^{-6} g(x) - f(x)\,dx + \int_{-6}^{0} f(x) - g(x)\,dx$

$\therefore$ **D** **[1 mark]**

3 $\int_0^4 (3\sqrt{x}+x)\,dx = \int_0^4 (3x^{\frac{1}{2}} + x)\,dx$

$= \left[\dfrac{3\times 2x^{\frac{3}{2}}}{3} + \dfrac{x^2}{2}\right]_0^4$

$= \left[2x^{\frac{3}{2}} + \dfrac{x^2}{2}\right]_0^4$

$= \left(2\times 4^{\frac{3}{2}} + \dfrac{4^2}{2}\right) - (0+0)$

$= 2\times 2^3 + 8$

$= 24$

$\therefore$ **A** **[1 mark]**

4 $\dfrac{d}{dx}e^{x^3+6x} = 3(x^2+2)\,e^{x^3+6x}$

So $\int 3(x^2+2)e^{x^3+6x}\,dx = e^{x^3+6x} + c$

$3\int (x^2+2)e^{x^3+6x}\,dx = e^{x^3+6x} + c$

So $\int (x^2+2)e^{x^3+6x}\,dx = \dfrac{1}{3}\int 3(x^2+2)e^{x^3+6x}\,dx$

$= \dfrac{1}{3}e^{x^3+6x} + c$

$\therefore$ **C** **[1 mark]**

5 Total change $= \int_a^b R'(t)\,dt$

$$= \int_0^5 10e^{0.2t}\,dt$$

$$= \left[\frac{10e^{0.2t}}{0.2}\right]_0^5$$

$$= \left[50e^{0.2t}\right]_0^5$$

$$= 50e^1 - 50e^0$$

$$= 85.914\ldots$$

$\therefore$ **B** **[1 mark]**

Part B – Short-answer questions

1 a $\int (3x^4 + 2x)dx = \int 3x^4\,dx + \int 2x\,dx$

$$= \frac{3x^5}{5} + \frac{2x^2}{2} + c$$

$$= \frac{3x^5}{5} + x^2 + c$$ **[1 mark]**

b $\int (4 - 3x)^2\,dx = \int (16 - 24x + 9x^2)dx$

$$= 16x - \frac{24x^2}{2} + \frac{9x^3}{3} + c$$

$$= 16x - 12x^2 + 3x^3 + c$$ **[1 mark]**

c $\int \frac{3x^2 - 4x + 7}{\sqrt{x}}dx = \int (3x^2 - 4x + 7)x^{-\frac{1}{2}}\,dx$

$$= \int (3x^{\frac{3}{2}} - 4x^{\frac{1}{2}} + 7x^{-\frac{1}{2}})\,dx$$ **[1 mark]**

$$= \frac{3x^{\frac{5}{2}}}{\frac{5}{2}} - \frac{4x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{7x^{\frac{1}{2}}}{\frac{1}{2}} + c$$

$$= \frac{2 \times 3x^{\frac{5}{2}}}{5} - \frac{2 \times 4x^{\frac{3}{2}}}{3} + 14x^{\frac{1}{2}} + c$$

$$= \frac{6x^2\sqrt{x}}{5} - \frac{8x\sqrt{x}}{3} + 14\sqrt{x} + c$$ **[1 mark]**

2 a $\int_{-1}^{2}(9x^2 - 6x + 2)dx = \left[\frac{9x^3}{3} - \frac{6x^2}{2} + 2x\right]_{-1}^{2}$

$= \left[3x^3 - 3x^2 + 2x\right]_{-1}^{2}$ **[1 mark]**

$= \left[3 \times 2^3 - 3 \times 2^2 + 2 \times 2\right] - \left[3 \times (-1)^3 - 3 \times (-1)^2 + 2 \times -1\right]$

$= 16 + 8$

$= 24$ **[1 mark]**

b $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 4\sin 3x\,dx = \left[-\frac{4\cos 3x}{3}\right]_{-\frac{\pi}{6}}^{\frac{\pi}{6}}$ **[1 mark]**

$= \left(-\frac{4\cos\frac{\pi}{2}}{3}\right) - \left(-\frac{4\cos -\frac{\pi}{2}}{3}\right)$

$= 0 - 0$

$= 0$ **[1 mark]**

c $\int_{1}^{3}\frac{dx}{(2x+5)^2} = \int_{1}^{3}(2x+5)^{-2}\,dx$

$= \left[\frac{(2x+5)^{-1}}{2 \times -1}\right]_{1}^{3}$

$= \left[\frac{-1}{2(2x+5)}\right]_{1}^{3}$ **[1 mark]**

$= -\frac{1}{22} - \left(-\frac{1}{14}\right)$

$= \frac{2}{77}$ **[1 mark]**

3 $\frac{dy}{dx} = 14x - 4$

$y = 7x^2 - 4x + c$ **[1 mark]**

$y = 8$ when $x = -1$, so $8 = 7 \times (-1)^2 - 4 \times -1 + c$

$8 = 11 + c$

$c = -3$ **[1 mark]**

$y = 7x^2 - 4x - 3$ **[1 mark]**

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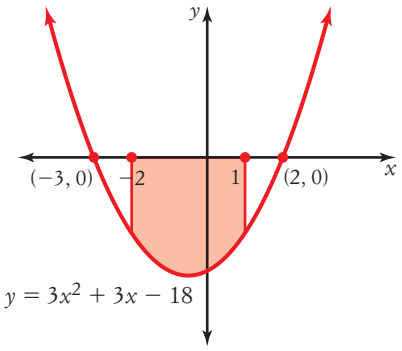
4 Draw a sketch of the parabola $y = 3x^2 + 3x - 18$.

Let $y = 0$: $3x^2 + 3x - 18 = 0$

$(3x - 6)(x + 3) = 0$

$x = 2$ or -3 **[1 mark]**

Zeros are located at $(-3, 0)$ and $(2, 0)$.



[1 mark]

Required area $= \int_{-2}^{1} (3x^2 + 3x - 18)dx$

$$= \left[\frac{3x^3}{3} + \frac{3x^2}{2} - 18x\right]_{-2}^{1}$$

$$= \left[x^3 + \frac{3x^2}{2} - 18x\right]_{-2}^{1}$$ **[1 mark]**

$$= \left(1^3 + \frac{3(1)^2}{2} - 18 \times 1\right) - \left((-2)^3 + \frac{3(-2)^2}{2} - 18 \times -2\right)$$

$$= -15\frac{1}{2} - 34$$

$$= -49\frac{1}{2}$$

The negative sign means the area is below the x-axis.

Area $= 49\frac{1}{2}$ units^2 **[1 mark]**

5 Solve simultaneous equations to find the intersection of the curve and line.

$y = x^2$ (1)

$y = x + 2$ (2)

$x^2 = x + 2$

$x^2 - x - 2 = 0$

$(x - 2)(x + 1) = 0$

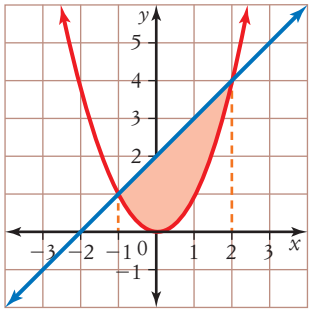
$x = 2$ or $x = -1$ **[1 mark]**

The curve $y = x^2$ is a parabola with a minimum at $(0, 0)$.

The line $y = x + 2$ has a positive slope and intersects the parabola at $x = 2$ and $x = -1$.

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Draw a sketch.



[1 mark]

$$\text{Area} = \int_{-1}^{2}(x+2)dx - \int_{-1}^{2}x^2\,dx$$

$$= \int_{-1}^{2}(x+2-x^2)dx$$ **[1 mark]**

$$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2}$$

$$= \left(\frac{2^2}{2} + 2\times 2 - \frac{2^3}{3}\right) - \left(\frac{(-1)^2}{2} + 2\times -1 - \frac{(-1)^3}{3}\right)$$

$$= \left(2 + 4 - \frac{8}{3}\right) - \left(\frac{1}{2} - 2 + \frac{1}{3}\right)$$

$$= 4\frac{1}{2}$$

Area between curves $= 4\frac{1}{2}$ units^2 **[1 mark]**

6 Total change $= \int_0^5 F'(t)dt$ **[1 mark]**

$$= \int_0^5 500e^{-0.4t}\,dt$$

$$= \left[\frac{500e^{-0.4t}}{-0.4}\right]_0^5$$

$$= \left[-1250e^{-0.4t}\right]_0^5$$ **[1 mark]**

$$= -1250e^{-2} + 1250e^{0}$$

$$= 1080.83\ldots$$

In the first 5 hours, about 1081 L flowed out of the tank. **[1 mark]**

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Part C – Analysis questions

1 Cross-sectional area = Area of rectangle + $\int_{-7}^{7}\left(\frac{4x^2}{25}-4\right)dx$ **[1 mark]**

$= \text{length} \times \text{width} + \left[\frac{4x^3}{25\times 3}-4x\right]_{-7}^{7}$

$= 14\times 6+\left(\frac{4\times 7^3}{75}-4\times 7\right)-\left(\frac{4\times(-7)^3}{75}-4\times -7\right)$ **[1 mark]**

$= 84+(-9.706\ldots\ -9.706\ldots)$

$= 64.586\ldots$ **[1 mark]**

Cross-sectional area $\approx 64.6\ \text{m}^2$

Volume = Area × length

$= 64.586\ldots \times 15$

$= 968.8\ \text{m}^3$ **[1 mark]**

2 $\frac{d}{dx}\left(\frac{1}{2}v^2\right)=6x^2-4x-3$

$\frac{1}{2}v^2=\int(6x^2-4x-3)dx$ **[1 mark]**

$=2x^3-2x^2-3x+c$ **[1 mark]**

When $x=0$, $v=3$

$\frac{1}{2}(3)^2=2(0)^3-2(0)^2-3(0)+c$

$c=\frac{9}{2}$ **[1 mark]**

$\frac{1}{2}v^2=2x^3-2x^2-3x+\frac{9}{2}$

$v^2=4x^3-4x^2-6x+9$

$v=\pm\sqrt{4x^3-4x^2-6x+9}$ **[1 mark]**

The condition $v=3$ when $x=0$ is satisfied by:

$v=\sqrt{4x^3-4x^2-6x+9}$ **[1 mark]**

When $x=2$,

$v=\sqrt{4(2)^3-4(2)^2-6(2)+9}$

$=\sqrt{13}$

When $x=2$, $v=\sqrt{13}$ **[1 mark]**

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